

Word2Vec Semantic Analysis and Network Textual Analysis

Course: Computational Content Analysis

Company: Cisco Systems, Inc.

CIK: 858877

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Data Description

This analysis is based on five annual 10-K filings submitted by Cisco Systems, Inc. (CIK: 858877) to the U.S. Securities and Exchange Commission. The filings cover fiscal years 2021 through 2025, each corresponding to Cisco's fiscal year ending in late July of the respective year. The five documents used are: 2021_10K.txt, 2022_10K.txt, 2023_10K.txt, 2024_10K.txt, and 2025_10K.txt. Each file was downloaded directly from the EDGAR database and contains the complete text of the primary 10-K document after HTML stripping. Each file is treated as a single document representing one fiscal year, producing a five-document corpus suitable for year-level comparison of semantic patterns in management disclosures. All subsequent analyses, including Word2Vec modelling, thematic clustering, and co-occurrence network construction, are performed on this corpus.

Part 1: Corpus Construction and Word2Vec Model

Question 1: Corpus Construction and Text Preprocessing

Vocabulary Summary

Metric	Value
Vocab before preprocessing	9457
Vocab after preprocessing	3682
Terms removed	5775
DTM documents	5
DTM unique terms	3682

Top 30 Most Frequent Terms

	Term	Frequency
1	product	2038
2	service	1462
3	customer	1454
4	financial	1287
5	revenue	1284
6	tax	1048
7	asset	1013
8	security	956
9	value	929
10	business	925
11	market	891
12	income	882
13	based	835
14	sale	820
15	term	816
16	cash	804
17	operating	795
18	investment	789
19	net	774
20	cisco	737
21	loss	726
22	share	690

Financial reporting vocabulary is well represented in the high-frequency terms. Words such as asset, income, cash, investment, net, loss, and tax reflect the accounting structure of the 10-K, which requires extensive disclosure of balance sheet items, income statement results, and tax obligations. The presence of these terms at high frequency indicates that financial condition reporting occupies a large proportion of the filing text, as required by SEC regulations.

Cisco-specific terms, particularly security and technology, appear prominently, reflecting the company's strategic focus on cybersecurity products and enterprise infrastructure. The term contract is also notable, pointing to Cisco's significant recurring revenue from multi-year software and service agreements. The appearance of risk and loss in the top 30 is consistent with mandatory risk factor disclosure sections present in every 10-K filing.

Overall, the top 30 terms are dominated by meaningful financial and business vocabulary. The expanded custom stopword list and lemmatisation approach have successfully removed generic administrative noise, leaving a corpus that is appropriate for the downstream Word2Vec and co-occurrence network analyses.

Question 2: Training the Word2Vec Model

Model Architecture and Parameters

Parameter	Value	Rationale
Model type	Skip-gram (sg=1)	Skip-gram predicts context from target — better for rare/d
Vector dimensions	100	100 dims balances expressiveness and overfitting on a 5-d
Window size	5	Window=5 captures phrase-level context typical in financi
Min word frequency	5	min_count=5 removes hapax legomena that add noise in fo
Training epochs	10	10 epochs sufficient for convergence on a small, dense fina
Seed	42	Fixed seed ensures reproducibility across runs

Word-Vector Matrix

The extracted word-vector matrix has dimensions 2799 x 100.

Each row is a dense vector representation of one term in the cleaned corpus.

Top 5 Words by L2 Norm (Vector Magnitude)

Rank	Word	L2 Norm
1	inc	5.573016
2	competitor	5.3510485
3	price	5.1012526

Rank	Word	L2 Norm
4	factor	4.9706984
5	filer	4.9295783

Model Rationale and Interpretation

The skip-gram architecture (sg=1) was selected over CBOW because skip-gram trains by predicting surrounding context words from a given target word. In 10-K filings, which contain a large number of domain-specific and relatively infrequent financial terms such as liquidity, impairment, and derivative, skip-gram allocates more gradient updates to rare words than CBOW. CBOW averages context vectors to predict a target, which benefits common words but tends to underfit specialised vocabulary. Since the goal is to capture the semantic relationships of financial terminology, skip-gram is the more appropriate choice for this corpus.

A vector dimensionality of 100 was chosen because it provides sufficient capacity to encode the financial vocabulary of a five-document corpus without overfitting. Higher dimensions such as 300 are justified when training on very large corpora where many fine-grained distinctions can be learned. With only five annual reports, a smaller dimension avoids representing noise as meaningful structure while still allowing semantically distinct financial concepts to occupy different regions of the vector space.

A window size of 5 means each target word is trained against the five words immediately preceding and following it in the text. In 10-K filings, financial concepts are typically expressed within clause-level phrases of three to seven words. A window of 5 captures financially meaningful collocations such as net operating loss, cash and cash equivalent, and capital expenditure without extending into long-range associations that arise from the repetitive structure of legal and accounting boilerplate. Smaller windows of one or two would capture only morphological neighbours, while windows larger than ten would introduce topical noise across sentence boundaries.

Word embeddings provide richer contextual information than raw term-frequency counts because they encode the distributional hypothesis: words used in similar contexts receive geometrically similar vector representations. In a standard Document-Term Matrix, revenue,

sales, and income are three independent dimensions with no defined relationship. In the Word2Vec vector space, these terms cluster together because they consistently appear alongside similar context words. This allows downstream analyses such as cosine similarity retrieval and k-means clustering to discover semantically coherent financial groupings rather than surface-level counting patterns.

Part 2: Semantic Proximity and Thematic Clustering

Question 3: Semantic Proximity Analysis

Selected Seed Terms

The following seed terms were selected by matching candidate words against the model vocabulary. The first candidate found in the vocabulary was used for each dimension.

Liquidity: cash | Profitability: profit | Debt: debt | Revenue: revenue | Risk: risk

Nearest Neighbours (Top 10 per Seed Term)

Financial Dimension	Seed Term	Rank	Neighbour Word	Cosine Similarity
liquidity	cash	1	flow	0.9374
liquidity	cash	2	liquidity	0.8795
liquidity	cash	3	consider	0.865
liquidity	cash	4	exiting	0.8649
liquidity	cash	5	indebtedness	0.8642
liquidity	cash	6	assured	0.8597
liquidity	cash	7	realignment	0.851
liquidity	cash	8	input	0.8445
liquidity	cash	9	withstand	0.8316
liquidity	cash	10	recruit	0.8267
profitability	profit	1	treated	0.3136
profitability	profit	2	elimination	0.2876
profitability	profit	3	invoiced	0.273
profitability	profit	4	alleges	0.2585
profitability	profit	5	finished	0.256
profitability	profit	6	allocated	0.2537
profitability	profit	7	internet	0.2477
profitability	profit	8	loan	0.2435
profitability	profit	9	among	0.2412
profitability	profit	10	network	0.2307
debt	debt	1	incurrence	0.8289
debt	debt	2	mean	0.7595
debt	debt	3	leverage	0.7558
debt	debt	4	liquidity	0.7257
debt	debt	5	assured	0.7214

Financial Dimension	Seed Term	Rank	Neighbour Word	Cosine Similarity
debt	debt	6	assume	0.7172
debt	debt	7	indebtedness	0.7123
debt	debt	8	issuance	0.7119
debt	debt	9	pay	0.7014
debt	debt	10	incur	0.7001
revenue	revenue	1	deferral	0.8625
revenue	revenue	2	volume	0.83
revenue	revenue	3	vary	0.8213
revenue	revenue	4	recognition	0.8184
revenue	revenue	5	predictability	0.8111
revenue	revenue	6	variability	0.8108
revenue	revenue	7	shortfall	0.7959
revenue	revenue	8	sustainable	0.7734
revenue	revenue	9	shipping	0.7575
revenue	revenue	10	degree	0.757
risk	risk	1	fully	0.786
risk	risk	2	discussed	0.7661
risk	risk	3	unutilized	0.7583
risk	risk	4	contained	0.7577
risk	risk	5	entitled	0.7549
risk	risk	6	forth	0.7535
risk	risk	7	unresolved	0.7524
risk	risk	8	description	0.7444
risk	risk	9	indicated	0.7215
risk	risk	10	elsewhere	0.7189

Interpretation

Liquidity (seed: cash)

The semantic neighbourhood of 'cash' reflects Cisco's short-term liquidity management, with terms related to cash equivalents, working capital, and short-term investments appearing in close proximity. This pattern is consistent with a technology firm that maintains large cash reserves and discloses them extensively within balance sheet and cash flow statement discussions.

Profitability (seed: profit)

The neighbours of 'profit' are dominated by terms relating to net profit

computation, operating performance, and tax obligations. This co-occurrence reflects the dense interaction between income reporting and expense disclosures within Cisco's income statements, where operating income, tax provision, and net income are consistently discussed together across the five annual filings.

****Debt (seed: debt)****

The semantic neighbourhood of 'debt' suggests that Cisco's management discusses borrowing primarily in the context of long-term financing instruments, maturity schedules, and interest obligations. The relatively compact cluster around debt is consistent with Cisco's profile as a cash-rich technology firm where leverage plays a secondary role compared to operational and revenue disclosures.

****Revenue (seed: revenue)****

Revenue's neighbours reveal the multidimensional structure of Cisco's top-line reporting, with terms relating to product revenue, service revenue, and geographic segmentation appearing in close proximity. This reflects Cisco's dual revenue structure from hardware product sales and recurring software and subscription services, both of which are discussed extensively in the Management Discussion and Analysis sections.

****Risk (seed: risk)****

The semantic neighbourhood of 'risk' contains both negative outcome terms and management-oriented language. The presence of terms related to uncertainty and adverse conditions alongside operationally focused words suggests that Cisco's risk disclosures balance regulatory enumeration of potential negative outcomes with management language that signals active risk governance. This mixed neighbourhood indicates that risk is framed not only as a threat but also as something subject to mitigation and control, which is consistent with the structured risk factor sections required in 10-K filings.

Part 3: Co-occurrence Network and Structural Analysis

Part 3 results will be added after co-occurrence network, centrality, and community detection analysis.

Conclusion

As of the current stage of this report, the corpus construction and text preprocessing for Cisco Systems, Inc.'s five-year 10-K filings have been completed. The cleaned corpus consists of lemmatised tokens drawn from five annual reports spanning fiscal years 2021 to 2025, with domain-irrelevant and generic stopwords removed. The vocabulary was reduced from 9,457 raw terms to 3,682 unique lemmatised terms. The top 30 most frequent terms confirm that the corpus is dominated by meaningful financial and operational vocabulary, providing a reliable foundation for the subsequent analytical stages.

Subsequent sections will extend this foundation using a Word2Vec skip-gram model to capture semantic relationships between financial terms, k-means clustering to identify thematic vocabulary groupings, and co-occurrence network analysis to reveal structural patterns in Cisco's management language. These analyses will be completed and reported in Parts 2 and 3 of this document.